

Automated Simplification of Clinical Discharge Summaries Using Large Language Models

Uziel Rivera-Lopez

Dr. Mehmet Dalkilic

April 14th, 2026

LUDDY

SCHOOL OF INFORMATICS,
COMPUTING, AND ENGINEERING

The Problem: Health Literacy Gap

The Clinical Note Problem

- Written by clinicians, for clinicians
- Dense medical jargon, non-standard acronyms, complex syntax
- Fragmented structure
- Not patient facing

Consequences for Patients

- Large % of patients cannot understand their own summaries
- Leads to medication errors and poor health outcomes
- Adds translation burden, in-office translators hard to access outside of office
- Barriers multiply by language, education and socioeconomic



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

Solution: Clinical Text Simplification via LLMs

Lexical

Replace complex medical terms with plain language ('myocardial infraction' to 'heart attack')

Syntactic

Break long sentences into shorter, clearer ones that follow natural reading patterns

Semantic

Remove low value clinical details that do not affect patient understanding or care decisions



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

What Has Been Tried

Lu et al.

Summary then simplify pipeline with narrative prompts. Outperformed BERT baseline by 0.49 pts on 5-pt scale. Reduced hallucinations via two-stage generation.

Jebich et al.

16 radiologists evaluated ChatGPT simplified radiology reports, but noted misinterpretation of medical terms, imprecise language, and residual hallucination.

Lyu et al.

138 chest CT + brain MRI reports simplified by ChatGPT, scored 4.2/5 — but only 37% of outputs gave specific actionable suggestions; the rest were too generic to help patients.

Fine-tuned mT5 (2023)

Fine-tuned mT5 outperformed GPT-3.5 (BLEU 42.6 vs 39.7). Domain adaptation on PubMed Central improved both fluency and factual preservation of drug names/diagnoses.



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

Bottlenecks & Technical Challenges

Hallucination/Factual

~7% of LLM outputs require clinician post-editing. Errors include wrong dosage, omitted allergies, and changed diagnoses - in general, can be harmful

Evaluation Standard

Human evaluation by clinicians remains the gold standard but is expensive and hard to scale. Automated consistency checks are an active research need.

Metric Insufficiency

Standard NLP metrics don't capture factual correctness or patient comprehension. Domain specific metrics are better but task specific and expensive

Generalization vs Specialization

General purpose LLMs (GPT) underperform fine-tuned models. Domain adaptation on biomedical text before fine tuning improves results



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

Datasets for the Pipelines

MIMIC-III / MIMIC-IV

Source: Beth Israel Deaconess Medical Center ICU

Type: De-identified HER data – discharge summaries, notes, labs

Access: PhysioNet (credential access)

Use: Source for discharge summaries and clinical notes pipeline

MTSamples

Source: mtsamples – public transcribed medical reports

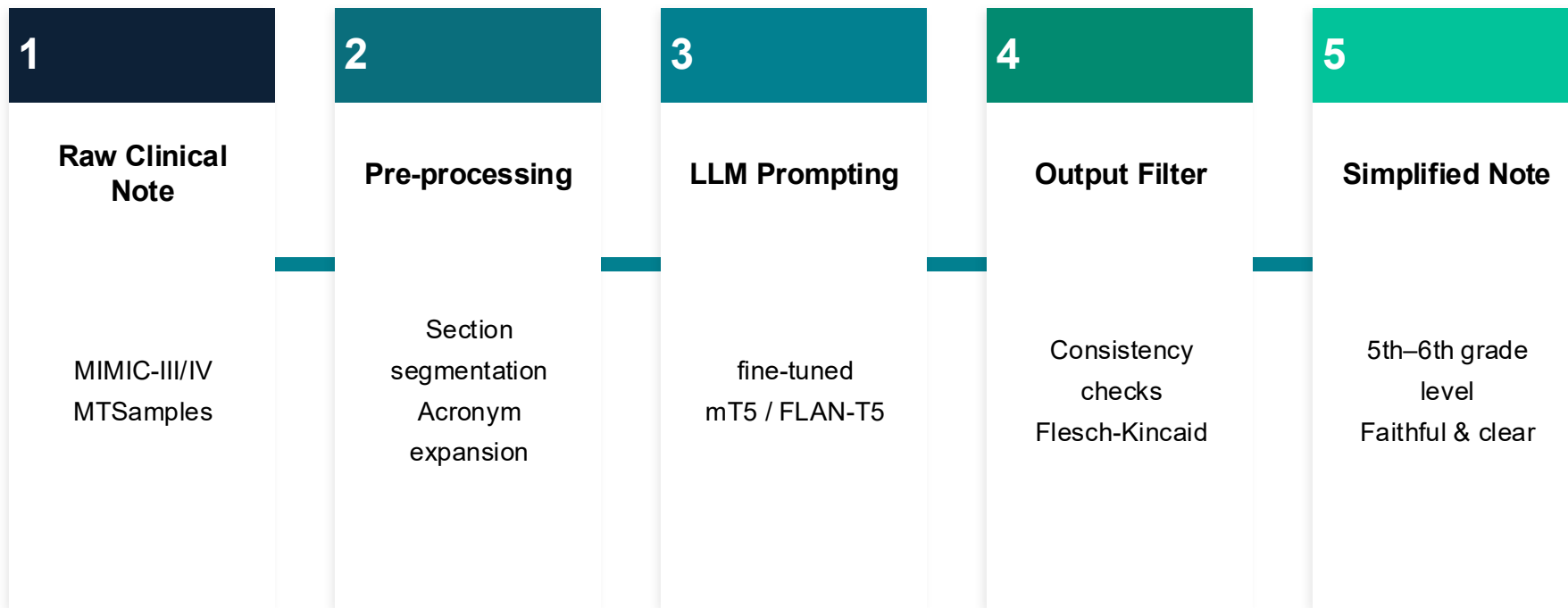
Type: Discharge summaries, consult notes, etc

Access: Open source, scrapped via custom python scrapper

Use: Fine tuning data; can be paired with MIMIC for coverage



Simplification Pipeline Architecture



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

Modeling Approach: Fine Tuning over Prompting

Why?

- Fine tuned mT5 > GPT
 - BLEU: 42.6 vs 39.7; Domain specific data matters
- Domain adaptation on PubMed Central before fine turning improves fluency and factual preservation
- LoRA/QLoRA allow fine tuning on a single GPU
- Produces a demonstrable, reproducible research artifact

Training Data Strategy

- Input-output pairs
 - Raw clinical note -> simplified version (6th grade constraint)
 - JSONL format for HuggingFace SFTTrainer / Unsloth
- Data augmentation
 - Paraphrase outputs
 - MTSamples: discharge + consult notes
- Target: 300-500 quality pairs for initial fine tune run



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

Completed

Literature Review

Cover clinical simplification, LLM 'faithfulness', pipelines, and evaluation metrics

Training Data

MTSamples scrapped with deduplications, MIMIC-III training is complete.

UI Mockups

Application prototype designed, and interface wired framed.

Model Selection & Fine Tuning

Evaluation FLAN-T5, BioMistral, mT5, LoRA / QLoRA setup in progress. Augmentation training pairs



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

What is next?

Immediate (Spring 2026)

- Finalize training data pairs (target 300–500)
- Run LoRA fine-tune on FLAN-T5 or BioMistral
- Compute section-level BLEU/ROUGE/FKGL scores
- Implement hallucination detection layer
- Complete full pipeline end-to-end demo

Future Research Directions

- Multilingual simplification (Spanish, Mandarin)
- Real-time clinical translation in EHR systems
- Patient comprehension user studies
- Larger model comparison (Llama 3, Mistral, BioGPT)



INDIANA UNIVERSITY

LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING